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## Input Scene



Familiar Video



EEG



fNIRS

New viewer (test participant)

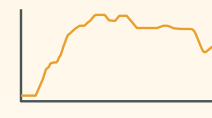
video time  $t$ 

2

## Two Parallel Branches

A

## Video-Time Prior (Low-Cost, Label-Derived)

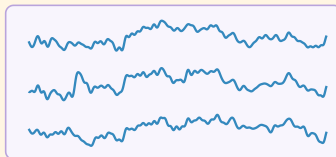
Training participants  
(labels available)Fold-wise  
median over  
participantsRadius-3  
within-video  
smoothingVideo-Time  
Prior  
 $\bar{p}_{v,t}$ 

Held-out participant labels excluded

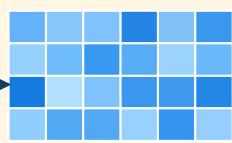
B

## EEG-fNIRS Branch (Participant-Specific)

EEG



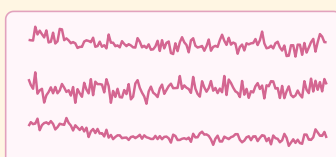
EEG features



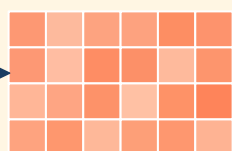
Graph Encoder

EEG Graph  
Encoder

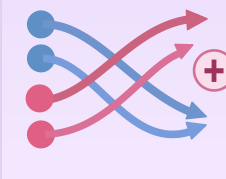
fNIRS



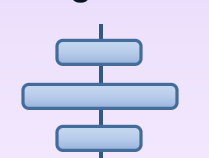
fNIRS features



Graph Encoder

fNIRS Graph  
EncoderCross-modal  
Attention

Regressor

Physiological  
Prediction  
 $m_{s,v,t}$ 

3

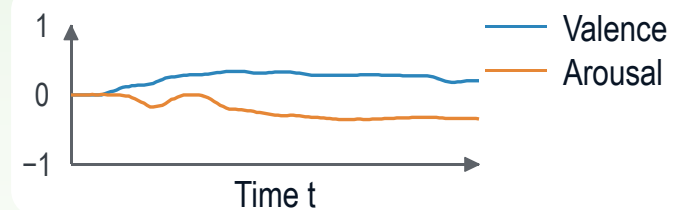
## Fusion and Output

Prior-Dominating  
Fixed Fusion

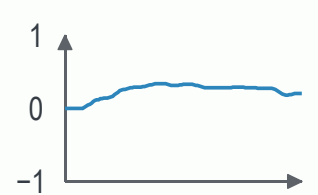
$$\hat{y} = w \odot \bar{p} + (1 - w) \odot m$$

$$w = [0.99, 0.92]$$

## Predicted affect trajectory

 $\hat{y}_{s,v,t}$ 

Valence over time



Arousal over time

